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Enriching type modification traits

Note: this is an early draft. It's known to be incomplet and incorrekt, and it has lots of bad fomattting.

Abstract

We introduce additional type traits to the standard library focused on type modification. The new type traits we present considerably simplify qualifiers manipulation. We also introduce a new type trait to remove all pointers on a type for the sake of completeness. These type traits have been especially useful in the design of proxy classes, included an updated design for bit manipulation utilities. They also have been used extensively in the implementation of a library dedicated to the creation of custom overload sets that will be proposed for standardization in a separate proposal.

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1 Proposal

[proposal]

1.1 History

[proposal.history]

- **P1450R2** targets LEWG. Changes required by LEWGI, and especially the removal of signedness and clone traits, have been implemented.
- **P1450R1** implemented the required changes with clone traits being marked for removal. The proposal was quickly reviewed by LEWGI at the Cologne 2019 Standards Committee Meeting to check that the changes required in the previous meeting were implemented. The proposal was then forwarded to LEWG with unanimous consent.
- **P1450R0** was carefully reviewed by LEWGI at the Kona 2019 Standards Committee Meeting, and was approved to be forwarded to LEWG with minor modifications. Jonathan Wakely and Ericnie Fiselier both confirmed that they have their own implementation of some of these type traits as an internal detail of their codebase. LEWGI recommended the removal of the `clone_*` form, as well as `copy_signedness`, and suggested the removal of `copy_all_extents` and `copy_all_pointers`. For consistency with the rest of the type traits, the last two are kept for now, and will be removed if LEWG wants to. LEWGI did not find a better name than `copy_*` to name these type traits.
- At the San Diego 2018 Standards Committee Meeting, LEWGI recommended the extraction of simple type modification traits from the original **P1016R0** proposal, which led to the proposal **P1450R0** in its current form.
- A first proposal was submitted in 2018 as **P1016R0**. However, back then it was not clear whether this kind of type traits should wait for reflection. In San Diego 2018, SG7 clarified that these type traits are pure library facilities that do not need to be first reviewed by them. They also clarified the fact that type traits in their current form live in a different space than reflection, and that the second one will not make the first one disappear. As a consequence, basic type traits should not wait for reflection. In San Diego, LEWGI recommended the extraction of simple type modification traits from the original **P1016R0** proposal, which led to this proposal in its current form.
- Originally developed as helper traits for an overload sets and sequences library, presented at CppCon 2018 (**Custom Overload Sets and Inline SFINAE for Truly Generic Interfaces**). Early discussions and feedback included a thread entitled **General purpose utilities for template metaprogramming and type manipulation** on the future proposals Google group.

1.2 Introduction

[proposal.introduction]

Since their introduction with C++11, the standard library type traits have been of great help for template metaprogramming. They contributed to the standardization of common metaprogramming patterns, such as SFINAE with `enable_if`, and since C++17 with `void_t`. In this paper, we introduce new type traits corresponding to metaprogramming patterns that turned out to be very useful to implement template proxy classes as well as to implement a tool to build custom overload sets. This tool will be proposed for standardization in a separate paper. We believe that the listed type traits are of common use and could benefit the entire community. The new type traits fall in three different categories:

- pointers removal: `remove_all_pointers` inspired from `remove_all_extents`
- qualifiers manipulation: `copy_*` type traits

An implementation is available at <https://github.com/vreverd/type-utilities>.

1.3 Impact on the standard

[proposal.impact]

This proposal is a pure library extension. It does not require changes to any standard classes or functions. All the extensions belong to the `<type_traits>` header.

1.4 Motivations and design decisions

[proposal.design]

1.4.1 Pointers removal

[proposal.design.removal]

```
// Pointers removal
template <class T> struct remove_all_pointers;

// Type alias
template <class T> using remove_all_pointers_t = typename remove_all_pointers<T>::type;
```

The current standard library includes two type traits to manipulate extents: `remove_extent` which removes the first array dimension, and `remove_all_extents` which removes all dimensions. For pointers, only one is currently provided: `remove_pointer` which removes one pointer. However, in some contexts it can be useful to access the “raw” type

However for the same reason that it can be useful to remove all dimensions, it can sometimes be useful to remove all pointers and access the “raw” type. Also, in the context of qualifiers manipulation (see (1.4.2)), it makes sense to provide tools to transform a `int***` into a `double***` by transferring all pointers from one type to another: `copy_all_pointers`. In this context, being able to remove all pointers seems to be a natural addition to the standard library, for completeness. For all these reasons, we propose to introduce the type trait: `remove_all_pointers`.

1.4.2 Qualifiers manipulation

[proposal.design.copy]

```
// Qualifiers manipulation
template <class From, class To> struct copy_const;
template <class From, class To> struct copy_volatile;
template <class From, class To> struct copy_cv;
template <class From, class To> struct copy_reference;
template <class From, class To> struct copy_extent;
template <class From, class To> struct copy_all_extents;
template <class From, class To> struct copy_pointer;
template <class From, class To> struct copy_all_pointers;
template <class From, class To> struct copy_cvref;

// Type aliases
template <class F, class T> using copy_const_t = typename copy_const<F, T>::type;
template <class F, class T> using copy_volatile_t = typename copy_volatile<F, T>::type;
template <class F, class T> using copy_cv_t = typename copy_cv<F, T>::type;
template <class F, class T> using copy_reference_t = typename copy_reference<F, T>::type;
template <class F, class T> using copy_extent_t = typename copy_extent<F, T>::type;
template <class F, class T> using copy_all_extents_t = typename copy_all_extents<F, T>::type;
template <class F, class T> using copy_pointer_t = typename copy_pointer<F, T>::type;
template <class F, class T> using copy_all_pointers_t = typename copy_all_pointers<F, T>::type;
template <class F, class T> using copy_cvref_t = typename copy_cvref<F, T>::type;
```

In the heavy template metaprogramming involved in the building of template proxy classes and custom overload sets, one pattern happened to be very useful: being able to transfer the qualifiers of one type to another one. For example, to transform a `const int&` into a `const double&`, a `int[1][2][3]` into a `double[1][2][3]`, or an `int***` to a `double***`. It can be also used in a function taking a universal reference as an input, to qualify another type based on the qualification of the input:

```
template <class T> void f(T&& x) {
    // An integer with the same qualification as the input
}
```

```

    using integer = std::copy_cvref_t<T&&, int>;
    /* function contents */
}

```

or to make a type `const` depending on another type:

```

template <class T> struct foo {
    // Data members
    T a;
    std::copy_const_t<T, int> n;
    std::copy_const_t<T, double> x;
    /* class contents */
};

```

Another uses are illustrated in [P0847R0](#), where `copy_cvref_t` is called `like_t`.

For completeness, qualifier manipulators are added to all existing categories of type transformations: `cv` ([2.1.7.1](#)), `reference` ([2.1.7.2](#)), `array` ([2.1.7.4](#)) and `pointer` ([2.1.7.5](#)). Additionally, depending on the behavior regarding the second template parameter, two kinds of qualifier parameters are introduced: the copiers `copy_*` presented in the next section.

The complete list of proposed `copy_*` traits is:

- const-volatile modifications: `copy_const`, `copy_volatile`, `copy_cv`
- reference modifications: `copy_reference`
- array modifications: `copy_extent`, `copy_all_extents`
- pointer modifications: `copy_pointer` `copy_all_pointers`
- other transformations: `copy_cvref`

As a note, in the same way `remove_pointer` deals with cv-qualified pointers, `copy_pointer` `copy_all_pointers` copy the cv-qualifiers of pointers.

1.5 Technical specification [proposal.spec]

See the wording (part [2](#)).

1.6 Discussion and open questions [proposal.discussion]

1.6.1 Bikeshedding [proposal.discussion.bikeshed]

While some names are straightforward and follow existing patterns in standard library, the following names are the most likely to be debated:

- `copy_*`

This name has been validated by LEWGI at the Kona 2019 meeting.

1.7 Acknowledgements [proposal.ackwldgmnts]

The authors would like to thank the participants to the related discussion on the [future-proposals](#) group. This work has been made possible thanks to the National Science Foundation through the awards CCF-1647432 and SI2-SSE-1642411.

1.8 References [proposal.references]

[A few additional type manipulation utilities](#), Vincent Reverdy, *Github* (March 2018)

[P1016R0](#), A few additional type manipulation utilities, Vincent Reverdy, *ISO/IEC JTC1/SC22/WG21* (May 2018)

[N4727](#), Working Draft, Standard for Programming Language C++, Richard Smith, *ISO/IEC JTC1/SC22/WG21* (February 2018)

[P0847R0](#), Deducing this, Gasper Azman et al., *ISO/IEC JTC1/SC22/WG21* (February 2018)

[General purpose utilities for template metaprogramming and type manipulation](#), ISO C++ Standard - Future Proposals, *Google Groups* (March 2018)

2 Wording

[wording]

2.1 Metaprogramming and type traits

[meta]

2.1.1 Requirements

[meta.rqmts]

¹ No modification.

2.1.2 Header `<type_traits>` synopsis

[meta.type.synop]

¹ Add the following to the synopsis of `<type_traits>`:

```
namespace std {
    // 2.1.3, helper classes

    // 2.1.4.1, primary type categories

    // 2.1.4.2, composite type categories

    // 2.1.4.3, type properties

    // 2.1.5, type property queries

    // 2.1.6, type relations

    // 2.1.7.1, const-volatile modifications
    template <class From, class To> struct copy_const;
    template <class From, class To> struct copy_volatile;
    template <class From, class To> struct copy_cv;

    template <class From, class To>
    using copy_const_t = typename copy_const<From, To>::type;
    template <class From, class To>
    using copy_volatile_t = typename copy_volatile<From, To>::type;
    template <class From, class To>
    using copy_cv_t = typename copy_cv<From, To>::type;

    // 2.1.7.2, reference modifications
    template <class From, class To> struct copy_reference;

    template <class From, class To>
    using copy_reference_t = typename copy_reference<From, To>::type;

    // 2.1.7.3, sign modifications

    // 2.1.7.4, array modifications
    template <class From, class To> struct copy_extent;
    template <class From, class To> struct copy_all_extents;

    template <class From, class To>
    using copy_extent_t = typename copy_extent<From, To>::type;
    template <class From, class To>
    using copy_all_extents_t = typename copy_all_extents<From, To>::type;
```

```

// 2.1.7.5, pointer modifications
template <class T> struct remove_all_pointers;
template <class From, class To> struct copy_pointer;
template <class From, class To> struct copy_all_pointers;

template <class T>
using remove_all_pointers_t = typename remove_all_pointers<T>::type;
template <class From, class To>
using copy_pointer_t = typename copy_pointer<From, To>::type;
template <class From, class To>
using copy_all_pointers_t = typename copy_all_pointers<From, To>::type;

// 2.1.7.6, other transformations
template <class From, class To> struct copy_cvref;

template <class From, class To>
using copy_cvref_t = typename copy_cvref<From, To>::type;

// 2.1.8, logical operator traits

// 2.1.9, endian
}

```

2.1.3 Helper classes

[meta.help]

¹ No modification.

2.1.4 Unary type traits

[meta.unary]

¹ No modification.

2.1.4.1 Primary type categories

[meta.unary.cat]

¹ No modification.

2.1.4.2 Composite type traits

[meta.unary.comp]

¹ No modification.

2.1.4.3 Type properties

[meta.unary.prop]

¹ No modification.

2.1.5 Type property queries

[meta.unary.prop.query]

¹ No modification.

2.1.6 Relationships between types

[meta.rel]

¹ No modification.

2.1.7 Transformations between types

[meta.trans]

2.1.7.1 Const-volatile modifications

[meta.trans.cv]

¹ Add the following to the table “Const-volatile modifications”:

Table 1 — Const-volatile modifications

Template	Comments
<code>template<class From, class To> struct copy_const;</code>	The member typedef <code>type</code> names the same type as <code>add_const_t<To></code> if <code>is_const_v<From></code> , and <code>To</code> otherwise.
<code>template<class From, class To> struct copy_volatile;</code>	The member typedef <code>type</code> names the same type as <code>add_volatile_t<To></code> if <code>is_volatile_v<From></code> , and <code>To</code> otherwise.
<code>template<class From, class To> struct copy_cv;</code>	The member typedef <code>type</code> names the same type as <code>copy_const_t<From, copy_volatile_t<From, To>></code> .

2.1.7.2 Reference modifications[\[meta.trans.ref\]](#)

- ¹ Add the following to the table “Reference modifications”:

Table 2 — Reference modifications

Template	Comments
<code>template<class From, class To> struct copy_reference;</code>	The member typedef <code>type</code> names the same type as <code>add_rvalue_reference_t<To></code> if <code>is_rvalue_reference_v<From></code> , <code>add_lvalue_reference_t<To></code> if <code>is_lvalue_reference_v<From></code> , and <code>To</code> otherwise.

2.1.7.3 Sign modifications[\[meta.trans.sign\]](#)

- ¹ No modification.

2.1.7.4 Array modifications[\[meta.trans.arr\]](#)

- ¹ Add the following to the table “Array modifications”:

Table 3 — Array modifications

Template	Comments
<code>template<class From, class To> struct copy_extent;</code>	The member typedef <code>type</code> names the same type as <code>To[extent_v<From>]</code> if <code>rank_v<From> > 0</code> && <code>extent_v<From> > 0</code> , <code>To[]</code> if <code>rank_v<From> > 0</code> && <code>extent_v<From> == 0</code> , and <code>To</code> otherwise. <i>Requires:</i> <code>To</code> shall not be an array of unknown bound along its first dimension if <code>From</code> is an array of unknown bound along its first dimension.
<code>template<class From, class To> struct copy_all_extents;</code>	The member typedef <code>type</code> names the same type as <code>copy_extent_t<From, copy_all_extents_t<std::remove_extent_t<From>, To>></code> if <code>rank_v<From> > 0</code> , and <code>To</code> otherwise. <i>Requires:</i> <code>From</code> and <code>To</code> shall not be arrays of unknown bounds along their first dimension at the same time.

2.1.7.5 Pointer modifications[\[meta.trans.ptr\]](#)

- ¹ Add the following to the table “Pointer modifications”:

Table 4 — Pointer modifications

Template	Comments
<pre>template<class T> struct remove_all_pointers;</pre>	The member typedef <code>type</code> names the same type as <code>remove_all_pointers_t<remove_pointer_t<T>></code> if <code>is_pointer_v<T></code> , and <code>T</code> otherwise.
<pre>template<class From, class To> struct copy_pointer;</pre>	The member typedef <code>type</code> names the same type as <code>copy_cv_t<From, add_pointer_t<To>></code> if <code>is_pointer_v<From></code> , and <code>To</code> otherwise.
<pre>template<class From, class To> struct copy_all_pointers;</pre>	The member typedef <code>type</code> names the same type as <code>copy_pointer_t<From, copy_all_pointers_t<std::remove_pointer_t<From>, To>></code> if <code>is_pointer_v<From></code> , and <code>To</code> otherwise.

2.1.7.6 Other transformations**[meta.trans.other]**

- ¹ Add the following to the table “Other transformations”:

Table 5 — Other transformations

Template	Comments
<pre>template<class From, class To> struct copy_cvref;</pre>	The member typedef <code>type</code> names the same type as <code>copy_reference_t<From, copy_cv_t<remove_reference_t<From>, To>></code> .

2.1.8 Logical operator traits**[meta.logical]**

- ¹ No modification.

2.1.9 Endian**[meta.endian]**

- ¹ No modification.